

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279004

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

JANAKIRAMAN; Kirupakar et al.

INTEGRATED FLIGHT DECK

Abstract

A system for an aircraft configured to: provide one or more of the sensed aircraft characteristics to a flight planning tool for processing by one or more features of the flight planning tool; receive a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; compare a current flight setting as determined based on the one or more sensed aircraft characteristics from the plurality of sensors with the received recommended flight setting; determine a potential operational efficiency based on the comparison between the received recommended flight setting and the current flight setting; and provide the potential operational efficiency to a user based on an identity of the user.

Inventors: JANAKIRAMAN; Kirupakar (Phoenix, AZ), KHAN; Kalimulla (Bangalore, IN), RAJENDRAN; Ramkumar (Madurai, IN)

Applicant: HONEYWELL INTERNATIONAL INC. (Charlotte, NC)

Family ID: 91381515

Appl. No.: 19/206282

Filed: May 13, 2025

Foreign Application Priority Data

IN 202211070893

Dec. 08, 2022

Related U.S. Application Data

parent US continuation 18174553 20230224 parent-grant-document US 12327478 child US 19206282

Publication Classification

Int. Cl.: G08G5/21 (20250101); **G06Q10/047** (20230101); **G06Q30/04** (20120101); **G08G5/32** (20250101)

U.S. Cl.:

CPC G08G5/21 (20250101); **G06Q10/047** (20130101); **G06Q30/04** (20130101); **G08G5/32** (20250101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. application Ser. No. 18/174,553, filed Feb. 24, 2023, which claims the benefit of and priority to Indian Provisional Patent Application No. 202211070893, filed Dec. 8, 2022, the entirety of each of which is incorporated by reference herein.

TECHNICAL FIELD

[0002] Various embodiments of the present disclosure relate generally to the field of control systems and, more particularly, to the field of control systems for aircraft.

BACKGROUND

[0003] Data analytic capabilities related to the avionics domain have been traditionally restricted to inferring insights from a post-flight analysis because real time data may be inaccessible during flights due to impossibilities of connection or prohibitive cost. This post-flight analysis can rely on recorded flight data that is limited due to storage restrictions and a lack of interconnected systems configured for generating data (e.g., systems unlike Aviation IoT systems). Further, the insights generated using such techniques can take days, weeks, or months to incorporate in routine operations and inefficiencies and anomalies may continue in flights between the generation of useful data and implementation of new programs and procedures based on the insights gleaned from such time-late data.

[0004] Even in connected aircraft systems, which may have in-flight connectivity and can provide such data to off-aircraft information technology systems during flight, there may be difficulties analyzing such data in real time due to the cost of transferring large amounts of data from the aircraft, latency and loss of connectivity during a flight, an indeterministic time response for certain real-time decision support systems, a lack of attribution/identification capabilities in on-board systems, and the like. The present disclosure is directed to overcoming one or more of these above-referenced challenges or shortfalls.

SUMMARY OF THE DISCLOSURE

[0005] According to certain aspects of the disclosure, methods and systems of providing real time insight using data collected from one or more systems integrated with a virtual flight deck are disclosed.

[0006] For instance, in one embodiment, a system for an aircraft includes a plurality of sensors configured to sense one or more aircraft characteristics of an aircraft system and generate a signal based on the sensed aircraft characteristic, a processing device communicatively coupled to the plurality of sensors, and a memory communicatively coupled to the processing device and storing processor-readable instructions that when executed by the processing device cause the system to: provide one or more of the sensed aircraft characteristics to a flight planning tool for processing by one or more features of the flight planning tool; receive a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; compare a current flight setting as determined based on the one or more sensed aircraft characteristics from the plurality of sensors with the received recommended flight setting; determine a potential operational efficiency

based on the comparison between the received recommended flight setting and the current flight setting; and display or transmit the potential operational efficiency to a user based on an identity of the user.

[0007] In another embodiment, a method includes: providing one or more sensed aircraft characteristics to a flight planning tool for processing by one or more features of the flight planning tool; receiving a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; comparing a current flight setting as determined by one or more of a plurality of sensors with the received recommended flight setting; determining a potential operational efficiency based on the comparison between the received recommended flight setting and the current flight setting; and providing the potential operational efficiency to a user based on an identity of the user.

[0008] In yet another embodiment, a method of operating an aircraft includes: providing one or more of sensed aircraft characteristics to a flight planning tool for processing by one or more features of the flight planning tool; receiving a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; comparing a current flight setting as determined by one or more of a plurality of sensors with the received recommended flight setting and automatically logging the current flight setting in a flight log; determining a potential operational efficiency based on the comparison between the received recommended flight setting and the current flight setting; and providing the potential operational efficiency to a user based on an identity of the user.

[0009] Additional objects and advantages of the disclosed embodiments will be set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practice of the disclosed embodiments.

[0010] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0012] FIG. 1 depicts an exemplary GUI for utilizing an integrated flight deck, according to one or more embodiments.

[0013] FIG. 2 depicts an exemplary diagram of a data flow for an exemplary aircraft-based network system.

[0014] FIG. 3A depicts a portion of a schematic block diagram of a framework of a platform of the networked computing system environment of FIG. 2.

[0015] FIG. 3B depicts a portion of a schematic block diagram of a framework of a platform of the networked computing system environment of FIG. 2.

[0016] FIG. 4 depicts an exemplary system architecture for providing insights using a flight management system.

[0017] FIG. 5 depicts a flow chart for providing insights using a flight management system.

[0018] FIG. 6 depicts exemplary key performance indicators provided using a flight management system.

[0019] FIG. 7A depicts further exemplary key performance indicators provided using a flight management system.

[0020] FIG. 7B depicts yet further exemplary key performance indicators provided using a flight

management system.

[0021] FIG. **8** depicts one or more events tracked using a flight management system for providing one or more key performance indicators.

[0022] FIG. **9** depicts exemplary charts for tracking one or more key performance indicators using a flight management system.

[0023] FIG. **10** depicts one or more key performance indicators for tracking with a flight management system.

[0024] FIG. **11** depicts additional key performance indicators for tracking with a flight management system.

[0025] FIG. **12** depicts an exemplary log of events for tracking one or more events using a flight management system.

[0026] FIG. **13** depicts a multilayer environment for developing one or more insights using a flight management system.

[0027] FIG. **14A** depicts a system environment for developing one or more insights using a flight management system.

[0028] FIG. **14B** further depicts a system environment for developing one or more insights using a flight management system.

[0029] FIG. **15** depicts a system environment for making one or more recommendations in real time based on mission events.

[0030] FIG. **16** depicts a method for generating one or more recommendations regarding a flight status.

DETAILED DESCRIPTION OF EMBODIMENTS

[0031] Various embodiments of the present disclosure relate generally to the field of generating insights based on a flight management system. While this disclosure describes the methods and systems with reference to aircraft, it should be appreciated that the present methods and systems may be applicable to various other vehicles.

[0032] FIG. **1** depicts an exemplary GUI **101** including an exemplary flight path **102** for a flight from a first airport **104** (e.g., New York's LaGuardia) to a second airport **106** (e.g., London's Heathrow). It is to be understood that these airports are merely exemplary choices and the flight could be from any takeoff location to any landing location. The flight path **102** includes a number of waypoints **102a**, **102b**, etc. The exemplary GUI **101** includes a chart **108** that shows a shortcuts section **110** and a savings section **112**. The shortcuts section **110** may include information related to a first shortcut **114** and a second shortcut **116**, which may be flight paths in between various waypoints of the original flight path **102**. If the shortcuts were utilized by the crew operating the flight, it would cause the savings shown in the savings section **112**. The savings could be based on, for example, distance, fuel, time, or other factors (e.g., dollars). Specifically, utilizing the first short cut **114** would save 9 nautical miles (Nm), 51 pounds (Lbs) of fuel, and 1.2 minutes from the originally path **102**. The second short cut would save 5 Nm, 28 lbs of fuel, and 0.9 minutes of time. This information could be displayed to a user of the flight management system to make one or more decisions regarding the flight path and the decisions made could be tracked and savings based on such decisions reported to a user of the flight management system.

[0033] FIG. **2** depicts exemplary avionics and related connected systems which may be connected in a data flow **100**, according to one or more embodiments. In step **101**, an operator and/or engineer may use “data integration” engines to develop rules and/or one or more applications (“Engineering”) using an integrated flight deck (IFD) **109**, which can be a cloud based dashboard and may be displayed, for example, using the PED **50** of FIG. **2**. In step **102**, a safety officer and/or flight data analysis may use an app engine to develop SDKs (“Advance Tech”) that configure one or more applications, such as, for example, self-contained mission managers, virtual electronic flight books, or cloned cockpit dashboards in the IFD **109**. In steps **112**, **113**, and **114**, a reusable app core **112**, reusable user interface framework **113**, and reusable data adapter **114**, respectively,

may pack metadata and send that metadata to edge hardware via IFD **109** (e.g., using an orchestrator of a networked computing system, as discussed more particularly herein). In step **105**, an onboard gateway (e.g., the avionics IoT gateway **105** of FIG. **2**) may deliver avionics data **107** to an edge analytics software development kit (SDK) associated with intelligent apps **108**. The avionics data **107** may be consumed data through the SDK of intelligent apps **108**, at step **105**. As used herein, avionics data can refer to data generated by one or more avionics systems including, for example, communications, navigation, display and management systems, aircraft flight control systems, fuel systems, collision avoidance (e.g., airborne and ground) systems, flight recorders, weather systems, and aircraft management systems. The onboard intelligent applications **108** may be configured to consume the avionics data **107** via the onboard avionics IoT gateway (step **105**) to deliver one or more inferences, suggestions, or recommended actions to pilot **110** and/or crew of the aircraft using the IFD **109** (e.g., one or more actions that the pilot can take or the aircraft can perform to result in changing a flight operation, such as flight operation savings. Flight operation savings can be directed towards safety, maintenance, weather related operations, fuel conservation, resource management, etc. An actionable update (e.g., a sync) may then be made to system **111**. In some embodiments, the actionable inferences, recommendations, and suggestions along with additional information may be provided to an electronic flight bag application and/or one or more external or internal applications by the aircraft data gateway.

[0034] FIG. **3A** and FIG. **3B** depict an exemplary high-level architecture of herein disclosed system **200**. Specifically, system **200** describes an example IFD platform configured to provide end-to-end real-time data acquisition (e.g., to provide real-time flight insights and maintenance alerting) with seamless access to data from multiple and at times different or varying data sources (e.g., data from avionics data sources) and flexibility to configure events or report fault events. System **200** includes advanced fault detection logic that can be incorporated to identify anomalies without the constraint of computational power and data access. In turn, real-time detection helps end users (e.g., flight crew) to perform self-contained mission management and/or to take corrective actions to prevent operational disruptions.

[0035] As shown in FIG. **3A** and FIG. **3B**, the overall architecture of system **200** is illustrated including the SDK components and corresponding framework. System **200** uses real-time accurate models and visual analytics to deliver one or more cloned dashboards and/or virtual EFBs locally and/or remotely, during ground operations, inflight, and in real-time to the end user (e.g., a pilot, flight crew, etc.). System **200** is extensible and portable for deployment in any cloud or data center environment for providing an enterprise-wide, top to bottom view, displaying the status of processes, assets, people, and safety. System **200** includes a collection of framework components, SDKs, UI libraries, and rules that specify how applications are developed from the platform and how additional SDKs and UI libraries can be added to the platform. Further, system **200** supports end-to-end capability to execute digital twins (e.g., a digital avionics engine) against process data and to translate the output into actionable changes to one or more avionics systems as detailed further below.

[0036] Computer systems of system **200** may include any type or quantity of one or more processors and one or more data storage devices including memory for storing and executing applications or software modules. In one embodiment, the processors and data storage devices are embodied in server-class hardware, such as enterprise-level servers. For example, the processors and data storage devices may include any type or combination of application servers, communication servers, web servers, super-computing servers, database servers, file servers, mail servers, proxy servers, and/virtual servers. Further, the one or more processors are configured to access the memory and execute processor-readable instructions, which when executed by the processors configures the processors to perform a plurality of functions of the system **200**.

[0037] The computer systems of system **200** are configured to receive and transmit data between other components to produce a desired technical result. In one embodiment, system **200** is

organized into a plurality of layers of a networked computing environment comprising a coordinator **212** and an orchestrator **220**. As detailed further below, components of coordinator **212** are in communication with components of orchestrator **220** via a network. The network may be any suitable network or combination of networks and may support any appropriate protocol suitable for communication of data to and from components of coordinator **212** and orchestrator **220** via the cloud and between various other components in the networked computing system **200**. The network may include a public network (e.g., the Internet), a private network (e.g., a network within an organization), or a combination of public and/or private networks. The network may be configured to provide communication between various components depicted in the various figures.

[0038] The network may include one or more networks that connect devices and/or components in the network layout to allow communication between the devices and/or components. For example, the network may be implemented as the Internet, a wireless network, a wired network (e.g., Ethernet), a local area network (LAN), a Wide Area Network (WANs), Bluetooth, Near Field Communication (NFC), or any other type of network that provides communications between one or more components of the network layout. In some embodiments, the network may be implemented using cellular networks, satellite, licensed radio, or a combination of cellular, satellite, licensed radio, and/or unlicensed radio networks.

[0039] System **200** may support multiple layers, such as, for example, a presentation layer **223**, an application layer **209**, a framework layer **225**, a platform layer **227**, and an avionics layer **229**, as shown. In Aerospace Internet of Things (AIOT) systems, AIoT providers may direct their focus toward connecting onboard avionics devices, extracting data from those devices, and sending the data to the cloud for analytics with the help of one or more onboard IoT gateways (e.g., gateway connector library of the avionics SDK factory **228a-n**, other gateway SDKs **236** of the avionics layer **229**, etc.). In one embodiment, the presentation layer **223** of system **200** includes a plurality of graphical user interfaces (GUI) of applications **202**, **204** (e.g., an umbrella application GUI), **206**, as well as avionics GUI templates **208**. The presentation layer **223** and application layer **209** can include the application associated with system **200** and features hosted therein, including the GUI for a user to interact with the application as well as the business logic of the application.

Additional UI features can be added at any time as detailed in the reference architecture

[0040] The application layer **209** can include one or more hosted applications **214**, **216**, **218** with corresponding logic and concrete data objects. In some aspects, applications **214**, **216**, **218** may perform business logic on data streams. Communication between components of the presentation layer **223** (e.g., application GUIs **202**, **204**, **206**) and components of application layer **209** (e.g., applications **214**, **216**, **218**) may be performed using coordinator **212** via the network, which may include use data and security communications.

[0041] A connected avionics system, such as that described herein, may have the data stream injected into gateways from multiple onboard avionics subsystems, such as FMS **30**, an enhanced ground proximity warning systems (EGPWS), a weather radar, a brakes systems (not depicted), etc. for data transfer to cloud-based analytics. In this respect, orchestrator **220** may be configured to push and pull information from applications **214**, **216**, **218**, for example, upon receipt of instructions from users of application GUIs **202**, **204**, **206**, **208**. In some aspects, hosted application **214** can push data objects from orchestrator **220** and/or pull data from data sources in communication with orchestrator **220** (e.g., avionics data such as FMS, 4D flight plan, ground- or aircraft-based traffic management systems, etc.). Orchestrator **220** in turn can be in communication with an object factory **222** of the framework layer **225**, which creates the interface for data objects.

[0042] The framework layer **225** can be communicatively coupled between the SDKs of platform layer **227** and the applications of the application layer **209** and function as a standard interface that all SDKs can expose. In some aspects, the framework layer **225** creates concrete instances of the SDKs, managing the number of objects created. The object factory **222** can be communicatively coupled to components of the platform layer **227**. For example, object factory **222** can be

configured to handle objects from the avionics SDK factory **228a-n**, user interface (UI) SDKs **230a-n**, and platform support services **232a-n**. Avionics SDK factory **228a-n** can include SDKs related to cFMS, RADAR, wheels and brakes, speech, connected FMS, RADAR, gateway connectors, and the like. UI SDKs **230a-n** can include maps widgets, pages widgets, forms widgets, and/or the like. Platform support services **232a-n** can include components related to license management, identity and access management, certificate management, launch darkly, cloud connectors, logging and application dynamics, support libraries and the like. In some aspects, platform support services **232a-n** can be communicably connected to gateway devices, such as connected gateway devices.

[0043] Platform layer **227** can contain all of shared SDKs of the system **200**, whereby the shared SDKs contain all the connectivity to external devices and any other content that can be shared across applications. Components of avionics SDK factory **228a-n** can include hosted SDKs for avionics and/or flight management such as Flight Management Systems, Enhanced Ground Proximity Warning System (EGPWS), Traffic Collision Avoidance System TCAS, Weather or RADAR Systems, electronic flight bag (EFB) devices, wheels, connected FMS library, speech models, gateway connector libraries, etc. It is understood that these hosted SDKs can be software entities which can be included to generically manage interfaces and connections, provide access to engines, and host other shareable features. Avionics SDK factory **228a-n** can be communicatively coupled to the avionics layer **229**, including components thereof. For example, avionics SDK factory **228a-n** can be configured to connect with the avionics layer **229** and receive components therefrom. In some aspects, content to be shared that does not exist within a predetermined SDK can be encapsulated within a SDK wrapper. Such SDKs can come prepacked within platform layer **227** and provide core capabilities that can be required by an application (e.g., data logging, connection to both avionics and the cloud, data management, security management, etc.).

[0044] Components of the avionics layer **229** can include SDKs of aircraft servers (e.g., ADG400 SDK **234**) as well as other gateway SDKs **236**. SDKs of aircraft servers can in turn be communicatively coupled (e.g., SDKs of aircraft servers can filter and convert complex avionics data into various streaming formats such as JavaScript Object Notation (JSON), Extensible Markup Language (XML), etc.) and transmit the converted avionics data to various client applications with avionics data processing service (ADPS) gateway **238**. ADPS gateway **238** can serve as a point of entry between system **200** as well as a point of entry for a group of micro-services, such as the connected FMS micro-services **240**, takeoff-and-landing data systems (TOLD) **242**, and/or the like. ADPS gateway **238** may be, for example, configured to be updated each time a new micro-service is added or removed in the connected FMS micro-services **240**. ADPS gateway **238** may be implemented as hardware, software, and/or a combination of hardware and software.

[0045] A key attribute of the IFD of system **200** is the ability to add additional content as plugins to the existing application, providing extensibility, flexibility, and isolation of application features and customs processing logic. Therefore, in addition to object factory **222**, a plugin manager can be communicatively coupled to orchestrator **220**, whereby plugin manager can be in communication with registration store. In particular, system **200** includes components of the platform layer **227** and orchestrator **220**, including object factory **222**, plugin SDK **332a-n** and new SDKs **334a-n** to plugin, as shown in FIG. 3A and FIG. 3B. In so doing, platform layer **227** (e.g., orchestrator **220**) in concert with coordinator **212** enables system **200** to provide an umbrella application that can be found in an app store and once installed enables the purchase and use of additional hosted apps as they become available. For example, newer applications can be readily added at the application layer **209** and/or presentation layer **223**.

[0046] One or more rules engines of the system may be highly customizable frameworks for complex event processing. The term “engine” as used in this disclosure is understood as a self-contained piece of business functionality with interfaces contained within a hosted SDK (e.g., a microservice). Some non-limiting examples of engines contemplated for use with system **200** and

related methods of this disclosure can include an Avionics Engine, a Flight Management Engine (FME), Takeoff and Landing Engine (TOLDE), a Navigation Database Engine, as well as higher order content as a flight plan comparator utility. This enables efficient data filtering where patterns of incoming data are dynamic in nature. It not only helps in reducing the quantity of data but also ensures the quality of data being pushed to the cloud.

[0047] The herein discussed rules engine(s) may be used as a framework for building event-based workflows. The rules engine may include one or more of three main components: a message component, e.g., any incoming event; a rules node; and/or a rules chain. The message may refer to incoming data from devices, device life-cycle event, REST API event, RPC request, etc. The rules node may be a function that would be executed on an incoming message. There are many different node types that can filter, transform, and/or execute some action on incoming message. The rules chain may be nodes relate to each other with relations, so the outbound message from rule node is sent to next connected rule nodes.

[0048] Below are some common use cases for configuring rule chains: data validation and modification for incoming telemetry or attributes before saving to the database; copy telemetry or attributes from devices to related assets so you can aggregate telemetry, e.g., data from multiple avionics subsystems can be aggregated in related asset; create/update/clear alarms based on defined conditions; trigger actions based on edge life-cycle events, e.g., create alerts if device is online/offline; load additional data required for processing, e.g., load temperature threshold value for a device that is defined in device's customer or tenant attribute; trigger REST API calls to external systems; raise alarms/alerts when complex event occurs and use attributes of other entities inside email template; and/or consider user preferences during event processing. In some aspects, messages transmitted from system **200**, such as triggers and/or alerts, can be configured for transmitting information to flight crew and ground controllers for maintenance preparations reducing the aircraft-on-ground (AOG) times. System **200** can also be configured to detect near misses can also be configured in the IFD to build a trend model for early detection of anomalies before faults or malfunctions occur increasing safety.

[0049] FIG. **4** displays a network environment **450** for generating one or more recommendations and determining whether the recommendations were followed by a crew of an aircraft. The network environment **450** includes a network server **452**, a database **454**, a flight management system **456**, a flight data system **458**, and a flight management computing device **460**, which may include an input/output device **462** (e.g., a touch screen).

[0050] The flight management system (FMS) **456** may be a specialized computer system that may automate various in-flight tasks. One function of the FMS **456** may be to manage the flight plan for any given flight. The FMS **456** may receive input from various sensors and systems (not shown), such as, for example, a GNSS system, an INS system, a radio navigation system, etc. to determine the aircraft's position. Using the sensors and systems, the FMS may guide the aircraft along a given flight plan (e.g., from New York to London, as shown in FIG. **1**). The FMS may be controlled and displayed, for example, through a Control Display Unit (CDU) **464** which CDU **464** may incorporate various input/output devices (e.g., a small screen and keyboard and/or touchscreen). The FMS **456** may send a flight plan for display to an Electronic Flight Instrument System (EFIS), Navigation Display (ND), or Multifunction Display (MFD), which systems and displays could be displayed on the input/output device **462** of the computing device **460**, for example. In some embodiments, the FMS **456** may include, as one example, the flight management computing device **460**, the control display unit **464**, and a cross talk bus.

[0051] Still referring to FIG. **4**, the flight data system **458** may include, for example, one or more gateways that are configured to receive data from one or more aircraft systems (e.g., an FMS, a radar system, an engine system, an auxiliary power unit, wheel systems, brake systems, navigation, collision avoidance, flight recorders, automatic traffic control systems, etc.) and sensors (e.g., flow sensors, pressure sensors, temperature sensors, altimeters, airspeed indicators, position sensors,

oxygen sensors, force sensors, vibration sensors, compasses, magnetometers, gyroscopes, altitude heading sensors, tachometers, etc.) to record data generated by the aircraft during flight. The flight data system **458** may include one or more modules configured to enact one or more of its processes. For example, the flight data system **458** may include a network server platform, a network server device manager, an action handler module, a domain analytic applications module, a data store, a cloud connector module, a rules engine module, a bus management module, a node agent module, a data ingress/egress module(s), an edge analytics software development kit (SDK), a data bus, and one or more software API modules (e.g., a python API, a Linux API, etc.) The flight data system **458** may acquire, record, download, and analyze data in order to make one or more recommendations to the operators of the aircraft in real time.

[0052] Still referring to FIG. **4**, the flight management computing device **460** may include one or more preprogrammed instructions (e.g., stored in a communicatively coupled database) which may cause the flight management computing device **460** to demonstrate one or more advantages of using insights generated by the flight data system **458** to a user of the system **450**, which insights may be generated based on one or more features subscribed to by the operators of the aircraft. Exemplary features are shown in an advantages chart **466** of FIG. **2** and may include, among other things, a shortcut adviser, an RTA, and optimal step cruises. In some embodiments, the advantages chart **466** may display whether or not a feature is subscribed to, a potential savings for using such a feature, a cost to use such a feature, as well as other aspects related to the feature.

[0053] FIG. **5** depicts an exemplary flowchart **300** for determining whether a user and/or a group of users have interacted with various features of the system **450** for recommending actions to the aircraft operators. The various features may be accessed via one or more subscription settings (as explained in greater detail herein) and the subscription settings of the user and the actual usage of the various features may be tracked and an optimal subscription setting and/or optimal features for the user may be developed based on the subscription settings and an actual usage of the features of the flight planning tool as described herein. Flowchart **300** depicts an exemplary method for determining such optimal features. At step **302**, an operator may start a session. At step **304**, the session log may be started to record one or more actions of the user(s) and/or the system during the session. Alternatively, at step **306**, the user can start the session without starting a session log.

[0054] At step **308**, a user may access a connected FMS flight plan API. The display at step **308** may include a relatively large number of sessions (e.g., 6 sessions) because, in general, a user (e.g., a pilot) may routinely access an FMS flight plan API to get from a takeoff location to a landing location. In some embodiments, the connected FMS flight plan API may be a default location for beginning use of a flight plan

[0055] At step **310**, a user may access a connected FMS state API, for example, to determine a current FMS state. At step **312**, a user may access a shortcut adviser feature, e.g., the shortcut advisory displayed to a user in the chart of FIG. **1**. Other features which may be viewable and editable by a user include, for example, a navigation view feature, a radar API, and one or more other features. Each time a user enters or exits a feature, either for another feature or when exiting the FMS system completely, the system may tally the various uses, keeping a complete tally of the type of interaction, its duration, and other information about the user's interaction with the system such that the system can accurately log which functions and features of the system the user is utilizing and accurately charge the user for using the subscribed features as will be described in greater detail herein.

[0056] At step **314**, the user may access a nav view feature, which may provide a user access to various navigational aids at an input/output device, for example, the number of times the user accesses the nav view feature may be recorded. At step **316**, the user may end a session after having accessed the cFMS aircraft state API at step **310**. At step **318** the user may access a cFMS flight plan upload tool API for uploading one or more flight plans and may go on to later access one or more other features. At step **320**, a user may access a shortcut advisor feature, which feature may

recommend one or more shortcuts or savings of aircraft resources. At step **322**, a user may access a cMFS aircraft state API and may go on to access one or more additional features at, for example, steps **324**, **326**, and **328**. At step **324**, the user may access the nav view feature. At step **326**, the user may access the connected radar API and at step **328**, the user may access a shortcut advisor feature. [0057] Referring to FIG. **6**, in some embodiments, the system may record the actual usage of the various features by users of the system and the various figures described herein depict some exemplary charts, graphs, and other graphics for generating value based on the actual usage of the system described herein. For example, the actual usage of a user may be tracked using one or more of a daily sessions per user, a session duration, a total number of events for a given time period, an average number of events, a user number, a count, and a count per user. FIG. **6** shows specifically a daily sessions per user and/or a session duration for a given user, for a given group of users, for an entire user set, etc. As shown in chart **402**, the daily sessions per user can be tracked over time to determine whether a user group is using features more often over a given period. The FMS can track different information, for example, a number of total sessions for a user group and an average number of sessions per day. In some embodiments, as shown in chart **404**, the session duration may be recorded and reported in order to best understand how long a typical use of the FMS lasts in order to optimize a user group's interaction with the FMS. For example, an average session duration can be calculated and reported or the number of individual sessions lasting for a given time period (e.g., 0-10 s, 10-30 s, 30 s-1 min, etc.) can be recorded and reported.

[0058] Referring to FIG. **7A** and FIG. **7B**, a total number of events **502** may be tracked and recorded and displayed in a chart **500**. What qualifies as an event may be customizable by a user of the system and one or more default lists of events may be programmed into the system. Default events may include, for example, interaction with a given application, closing of a given application, opening of a given application, etc. Hence, the user's interaction with the system can be more accurately tracked and monitored in order to best value the use of the various systems which the user(s) utilize to generate events. In some embodiments, an average number events per a given time period can be tracked (e.g., average number of events per day for a given number of days, etc.) As shown by the differences between FIG. **7A** and FIG. **7B**, the graphics displayed to a user may be customizable.

[0059] Referring to FIG. **8**, in some embodiments, the system may be used to generate one or more searchable lists of events **602** and to display information **604** related to such searchable list of events **602**. The displayed information may be fileable and sortable based on, for example, a magnitude of the data. As shown in FIG. **6**, each time a user accesses one of the features (e.g., the "Nav View Feature," the "ATCT feature," "the cRadar API," etc.), the system may track a count (e.g., "29," "26," "24," etc.) The system may further count a number of accesses over a given period of time to determine a trend in the accessing of the feature. Further, the system can be configured to determine how many users actually access or changed a feature and deliver number of accesses per accessing user. Referring to FIG. **8** and FIG. **9**, the information displayed in FIG. **8** can be graphically depicted in order to best display the information to a user of the system.

[0060] Referring to FIG. **10**, the system may be configurable to track and analyze one or more system crashes, system errors, and/or system symbolization. A system user may utilize the number of crashes, errors, and/or system symbolization, for example, to determine an overall health of the system in order to debug the system and/or to determine what changes need to be made to the system. The number of crashes and/or system errors can be tracked over time and one or more tickets may be tracked to solve one or more of the crashes and/or system errors. Referring to FIG. **11**, the system may include one or more report trackers and/or affected user trackers. In some embodiments, the system may track a most affected device and/or most affected operating system in order to determine with which devices and operating system the system works best/worst.

[0061] Referring to FIG. **12**, an exemplary log of events for tracking one or more events using a flight management system is shown. More specifically, FIG. **12** shows the process level details of

various applications described herein, including the tasks running in parallel and their status at the time of a potential event. FIG. 12 provides various details about a call stack and a background thread of information during the event. Using the information described and shown in FIG. 12, a user (e.g., a developer) could conduct a root cause analysis or other type of analysis of a given event, tracking the various lines of software as they occur (e.g., are executed) in real time.

[0062] Referring to FIG. 13, a system diagram **1100** demonstrates one possible embodiment of a system hierarchy for generating in-flight insights and immediately demonstrating the value of such generated in-flight insights to a user. The system diagram **1100** includes an edge layer **1102**, a fog layer **1104**, and a cloud layer **1106**. At the edge layer **1102**, the system may include an FMS **1108**, one or more aircraft sensing systems such as a radar system **1110**, one or more engine systems **1112**, and an enhanced ground proximity warning system (EGPWS) **1114**. At the fog layer **1104**, the system may include a cloud connector module **1116**, an edge analytics module **1118**, an instrumentation module **1120** (which may receive input from a live telemetry module **1122**), and one or more business function modules **1124** (i.e., features) including, for example, a shortcut advisor feature **1126**, a weather hazard advisor feature **1128**, a flight level advisor feature **1130**, an emergency diversion feature **1132**, and one or more other features **1134**. The various features may provide one or more recommended flight settings that include one or more recommendations based on, for example, a flight path, a weather condition, an emergent diversion, and an optimal altitude. At the cloud layer **1106**, the system may include a data lake **1136**, one or more data analysis algorithms **1137**, one or more machine learning models **1138** that may generate one or more insights, and one or more configuration files **1140** for configuring the various data received through the fog layer **1104**. As shown in FIG. 13, in some embodiments, the plurality of sensors may be at an edge layer and the flight planning tool may be at a fog layer

[0063] FIG. 14A and FIG. 14B show an exemplary system architecture **1200** for constructing the system for generating insights based on real time data described herein. The system architecture includes a platform services module **1202**, a data access layer **1204**, an application layer **1206**, a platform software development kit (SDK) **1208**, a usage analytics agent **1210**, a public API **1212**, and a network server interface **1214**. The platform SDK **1208** may include, for example, a fog manager, a fog service mesh, a fog instrumentation service, one or more platform services, connector components connecting to one or more avionics systems, connector components for connecting to one or more external cloud environments, domain-specific analytics engines (e.g., Energy Management Engine, Shortcut Engine, FME, TOLDE, SDKs, platform services like identity and access management, license management, etc.) The usage analytics agent **1210** may keep track of all features (both subscribed and unsubscribed) and generate usage insights for the users of the system.

[0064] More specifically, FIG. 14A depicts components necessary to perform usage analytics of the various applications and features described herein. At one or more of the edge devices in the system described herein, it may be necessary to run, for example, a user analytics agent. The user analytics agent may monitor usage of the various features and post live telemetry information to an external server (e.g., a cloud server such as, for example, the Forge Cloud server). In some embodiments, the user analytics agent may post live telemetry information to the external server through the platform described herein. Further, at the external server (e.g., the cloud server) a data hive may capture live telemetry coming from the various edge devices. The data from the edge devices can be further processed by other applications to generate business intelligence (BI) reports to gain insights on the features that are used (e.g., the critical features, most/least utilized features, etc.) Such information can be used to provide the most useful features to a user of the system (e.g., a client).

[0065] FIG. 14B depicts various features and software modules running at the different software layers of the system (e.g., the application layer, data access layer, and platform services layer). In use, the user analytics agent can connect with any of these layers to send the usage information to

an external server.

[0066] Referring to FIG. 15, a system **1300** for a logging service hosted in an external flight planning tool that can be configured to customize data logging is shown. The system includes a configurable database **1302** that may be used to log one or more mission events **1304** and/or one or more data entry prompts **1306**. The mission events **1304** and the data entry prompts may be stored with metadata **1308**, **1310** in a mission event database **1311**. The system **1300** further includes an application **1312** (e.g., software) for logging one or more events and the application **1312** is communicatively coupled to the avionics hardware and software **1314** on the aircraft. The application **1312** may communicatively couple with onboard avionics systems and sensors and with a cloud-based infrastructure, which may be used to, for example, conduct post-mission analysis (e.g., in an application layer **1316**, a data processing layer **1318**, a network layer **1320**, and a sensing layer **1322**).

[0067] The system **1300** includes the configurable database **1302**, which may contain one or more trigger events (or “mission events”), and corresponding data that may be extracted from avionics and sensor systems (e.g., flight recorder, FMS, etc.) and temporarily stored in the external flight planning tool, which may be housed, for example, in the application **1312** and/or in the application layer **1316** of the cloud-based infrastructure. In embodiments, manual data entries, which may be required for logging, can be input based on policy or regulatory documents and the system **1300** may prompt entry of such entries and/or provide a form such that the application prompts a user to receive manual entries that will then be added to the logs. For example, in a situation in which an aircraft is making a transoceanic flight (e.g., as shown in FIG. 1), mission events like sequences of legs, coast-in, coast-out, equitime point, point of safe return, etc. can be configured and when triggered, the state of the aircraft at that point of time can be logged using a system such as that described with respect to FIG. 1. Manual entry triggers in the case of a transoceanic flight log could include entries during pre-flight, coast-in, coast-out and post-flight to record altimeter settings, navigation accuracy checks from, for example, both pilot and co-pilot sides of a cockpit etc. such that manually-recorded data entries during the transoceanic flight are manually logged and may be maintained on board the aircraft.

[0068] Still referring to FIG. 15, in some embodiments, based on the configured database, applications associated therewith may set the data access periodicity or subscribe to particular avionics-based system event to receive the data that is necessary to log. In embodiments, in order to minimize a data transfer to/from the aircraft, the data may be temporarily stored with a timestamp until the end of the flight or until crew manually requests archiving of the data on the cloud. Upon completion of the flight or upon a specific request from the crew, for example, the logged flight data may be sent to the cloud with appropriate labeling (e.g., aircraft registration, flight number, crew details, etc.) for retention in an off-aircraft database for a configured period of time.

[0069] Referring now to FIG. 16, a method of providing a potential operational flight efficiency to a user is shown. It is to be understood that the method shown in FIG. 16 is exemplary and additional, fewer, and/or other steps may be incorporated to the method without departing from the scope of the current claims.

[0070] At step **1402**, one or more of sensed aircraft characteristics may be provided to a flight planning tool for processing by one or more features of the flight planning tool. The sensed aircraft characteristics may be sensed, for example, by one or more of one or more of flow sensors, pressure sensors, temperature sensors, altimeters, airspeed indicators, position sensors, oxygen sensors, force sensors, vibration sensors, compasses, magnetometers, gyroscopes, altitude heading sensors, tachometers, navigation, collision avoidance, flight recorders, and automatic traffic control systems. The flight planning tool may be on board the aircraft and/or external to the aircraft and the aircraft may comprise one or more communications systems for communicatively coupling with the flight planning tool.

[0071] At step **1404**, one or more systems aboard the aircraft may receive a recommended flight

setting from the flight planning tool based on the processing by the flight planning tool. The recommended setting could include, for example, one or more recommended navigation changes (e.g., changes to one or more waypoints, etc.), one or more recommended altitude changes (e.g., one or more altitude step settings, etc), one or more aircraft settings changes (e.g., an engine speed, a system configuration, etc.) or other settings related to aircraft navigation and operational efficiency.

[0072] At step **1406**, a current flight setting, as determined based on the one or more sensed aircraft characteristics from the plurality of sensors, may be compared with the received recommended flight setting. For example, an aircraft heading (e.g., a current heading, a heading from one waypoint to another, etc.) may be compared with a recommended aircraft heading as received from the flight planning tool. In some embodiments, the current flight setting may be automatically logged at the time it is determined based on the one or more sensed aircraft characteristics. For example, in some instances, logs for flights (e.g., transoceanic flights) may require the verification of a “next waypoint” and “next to next waypoint” prior to reaching any particular waypoint along a planned route. In many instance, this has to be done a certain amount of time before the particular waypoint (e.g., 2 min prior to the waypoint). The logging of this event can be configured with associated data (outbound magnetic course and distance to the next waypoint presented in the flight management computer and verified for agreement with the planned course). Further, the flight management computer and other associated systems may be used to verify that, for example, after 10 minutes of having passed a waypoint, there is no deviation from a programmed route of flight. The system could verify this lack of deviation, at least in part by, for example, checking for cross track error. As another example, the system may use a steering long-range navigation system (“LRNS”) to verify a “TO” waypoint is consistent with a currently-active route clearance.

[0073] At step **1408**, a potential operational efficiency may be determined based on the comparison between the received recommended flight setting and the current flight setting. For example, if the recommended heading would save a certain amount of fuel when traveling from one waypoint to another (e.g., as determined by a flight management system) and would not lead to any collision or danger with other aerial objects (e.g., as determined by an air traffic control system) then this would be an operational efficiency (i.e., the saving of the fuel cost, the labor cost associated with adding additional fuel, etc.). If the recommended heading would also reduce flight distance/time and cause the aircraft to land earlier, then this would also potentially be an operational efficiency as it might save crew rest, avoid passenger delays, reduce engine and system running hours, etc. However, to continue with the example, if the alternate heading would reduce a flight distance but not save fuel and/or time (e.g., if a wind direction/speed caused flying at a particular heading to be less fuel/time efficient) or would result in other inefficiencies (e.g., there would be other flights landing at the hypothetical new landing time thus preventing the subject flight from landing and requiring circling the target airport, etc.) then changing a heading would not result in an operational efficiency and the change would not be recommended.

[0074] At step **1410**, the system may provide the potential operational efficiency to a user based on an identity of the user. For example, if the user is logged in and has a profile that is authorized to receive operational efficiencies as determined by, for example, an identity and access management system, then the system may provide the potential operational efficiency to the user.

[0075] At step **1412**, the system may convert the potential operational efficiency to a monetary value and provide the monetary value to a user. The monetary value may be based on, for example, the cost of fuel saved if the operational efficiency is enacted, the amount of man-hours and/or engine-hours saved if the operational efficiency is enacted, or based on some other factor. The monetary value for potential savings may be provided to a user even if the user is not using a particular application such as to receive recommendations from the application. That is, the monetary value may be provided as an enticement to the user to subscribe to use the application. Accordingly the identity (i.e., user profile) of the user may include one or more subscription

settings and the recommended flight setting and/or potential operational efficiency may be provided to the user based on the subscription settings of the user. The subscription settings could be stored, for example in a user profile module.

[0076] At step **1414**, the system may generate a bill, payable by the user or someone associated therewith, based on the converted monetary value of the potential operational efficiency. The generated bill may be a fraction of the monetary value of the potential operational efficiency. For example, if the system recommends an altered course that results in a savings of 50 lbs of fuel, the system may automatically convert the 50 lbs of fuel to a monetary value and present a bill to the user based on the cost of the savings. The conversion can be calculated in multiple ways. For example, the conversion could be based on a current price of fuel at a particular location (e.g., at the airport where the flight is to land, etc.), the conversion could be based on a pre-determined fuel price, the conversion could be based on an average fuel price, or other factors. The fraction of the saved money which is used to calculate the bill could be agreed upon by the parties and automatically supplied by the system. For example, the provider of the hardware and software that includes the flight planning tool could agree with the user that the user will pay 10%, 25%, 50%, etc. of the savings that are generated using the system to the provider.

[0077] It should now be understood that insights based on data generated by avionics systems and other on-board systems can be used to generate real-time actionable insights during flight preparations and during a flight and that such insights can be provided to a pilot and/or aircraft crew based on a subscription service subscribed to by the crew. In some embodiments, benefits of the subscription service can be provided to the crew in order to inform the crew how the benefits could increase efficiency and reduce costs associated with aviation operations.

[0078] The general discussion of this disclosure provides a brief, general description of a suitable computing environment in which the present disclosure may be implemented. In one embodiment, any of the disclosed systems, methods, and/or graphical user interfaces may be executed by or implemented by a computing system consistent with or similar to that depicted and/or explained in this disclosure. Although not required, aspects of the present disclosure are described in the context of computer-executable instructions, such as routines executed by a data processing device, e.g., a server computer, wireless device, and/or personal computer. Those skilled in the relevant art will appreciate that aspects of the present disclosure can be practiced with other communications, data processing, or computer system configurations, including: Internet appliances, hand-held devices (including personal digital assistants (“PDAs”)), wearable computers, all manner of cellular or mobile phones (including Voice over IP (“VoIP”) phones), dumb terminals, media players, gaming devices, virtual reality devices, multi-processor systems, microprocessor-based or programmable consumer electronics, set-top boxes, network PCs, mini-computers, mainframe computers, and the like. Indeed, the terms “computer,” “server,” and the like, are generally used interchangeably herein, and refer to any of the above devices and systems, as well as any data processor.

[0079] Aspects of the present disclosure may be embodied in a special purpose computer and/or data processor that is specifically programmed, configured, and/or constructed to perform one or more of the computer-executable instructions explained in detail herein. While aspects of the present disclosure, such as certain functions, are described as being performed exclusively on a single device, the present disclosure may also be practiced in distributed environments where functions or modules are shared among disparate processing devices, which are linked through a communications network, such as a Local Area Network (“LAN”), Wide Area Network (“WAN”), and/or the Internet. Similarly, techniques presented herein as involving multiple devices may be implemented in a single device. In a distributed computing environment, program modules may be located in both local and/or remote memory storage devices.

[0080] Aspects of the present disclosure may be stored and/or distributed on non-transitory computer-readable media, including magnetically or optically readable computer discs, hard-wired or preprogrammed chips (e.g., EEPROM semiconductor chips), nanotechnology memory,

biological memory, or other data storage media. Alternatively, computer implemented instructions, data structures, screen displays, and other data under aspects of the present disclosure may be distributed over the Internet and/or over other networks (including wireless networks), on a propagated signal on a propagation medium (e.g., an electromagnetic wave(s), a sound wave, etc.) over a period of time, and/or they may be provided on any analog or digital network (packet switched, circuit switched, or other scheme).

[0081] Program aspects of the technology may be thought of as “products” or “articles of manufacture” typically in the form of executable code and/or associated data that is carried on or embodied in a type of machine-readable medium. “Storage” type media include any or all of the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for the software programming. All or portions of the software may at times be communicated through the Internet or various other telecommunication networks. Such communications, for example, may enable loading of the software from one computer or processor into another, for example, from a management server or host computer of the mobile communication network into the computer platform of a server and/or from a server to the mobile device. Thus, another type of media that may bear the software elements includes optical, electrical and electromagnetic waves, such as used across physical interfaces between local devices, through wired and optical landline networks and over various air-links. The physical elements that carry such waves, such as wired or wireless links, optical links, or the like, also may be considered as media bearing the software. As used herein, unless restricted to non-transitory, tangible “storage” media, terms such as computer or machine “readable medium” refer to any medium that participates in providing instructions to a processor for execution.

[0082] The terminology used above may be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific examples of the present disclosure. Indeed, certain terms may even be emphasized above; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section. Both the foregoing general description and the detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed.

[0083] As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus.

[0084] In this disclosure, relative terms, such as, for example, “about,” “substantially,” “generally,” and “approximately” are used to indicate a possible variation of $\pm 10\%$ in a stated value.

[0085] The term “exemplary” is used in the sense of “example” rather than “ideal.” As used herein, the singular forms “a,” “an,” and “the” include plural reference unless the context dictates otherwise.

[0086] Other embodiments of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit of the invention being indicated by the following claims.

Claims

1. A system comprising: one or more sensors configured to sense one or more aircraft characteristics of an aircraft system; a processing device communicatively coupled to the one or more sensors; and a memory communicatively coupled to the processing device and storing

processor readable instructions that when executed by the processing device cause the system to: provide at least one aircraft characteristic of the one or more aircraft characteristics to a flight planning tool for processing by one or more features of the flight planning tool; receive a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; compare the recommended flight setting with a current flight setting that is determined based on at least one of the one or more aircraft characteristics from the one or more sensors; determine a potential operational efficiency based on the comparison between the recommended flight setting and the current flight setting; and change at least one operation of an aircraft based on the potential operational efficiency.

2. The system of claim 1, wherein the recommended flight setting comprises a recommendation that is determined based on at least one of a flight path of the aircraft or a weather condition in proximity of the aircraft.

3. The system of claim 1, wherein the one or more sensors are located at an edge layer onboard the aircraft and the flight planning tool is located at a fog layer onboard the aircraft.

4. The system of claim 1, wherein the flight planning tool comprises a flight management system.

5. The system of claim 1, wherein the potential operational efficiency is determined when the aircraft is in flight.

6. The system of claim 1, wherein the potential operational efficiency is determined when the flight planning tool is impacted by a loss of connectivity with a cloud-based device.

7. The system of claim 1, wherein the at least one operation of the aircraft comprises a navigation operation.

8. The system of claim 1, wherein the processor readable instructions, when executed by the processing device, further cause the system to: convert the potential operational efficiency to a value and provide the value to a user.

9. The system of claim 8, wherein the processor readable instructions, when executed by the processing device, further cause the system to: generate a bill, payable by the user, based on the value of the potential operational efficiency, wherein: the bill corresponds to a fraction of the value of the potential operational efficiency.

10. The system of claim 1, wherein the memory further includes processor-readable instructions that when executed by the processing device cause the system to: transmit the potential operational efficiency to a cloud-based device while the aircraft is in flight.

11. A method comprising: providing at least one aircraft characteristic of one or more aircraft characteristics sensed by one or more sensors to a flight planning tool for processing by one or more features of the flight planning tool; receiving a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; comparing the recommended flight setting with a current flight setting that is determined based on at least one of the one or more aircraft characteristics from the one or more sensors; determining a potential operational efficiency based on the comparison between the recommended flight setting and the current flight setting; and changing at least one operation of an aircraft based on the potential operational efficiency.

12. The method of claim 11, wherein the recommended flight setting comprises a recommendation that is determined based on at least one of a flight path of the aircraft or a weather condition in proximity of the aircraft.

13. The method of claim 11, wherein the one or more sensors are located at an edge layer onboard the aircraft and the flight planning tool is located at a fog layer onboard the aircraft.

14. The method of claim 11, wherein the potential operational efficiency is determined when the aircraft is in flight.

15. The method of claim 11, wherein the potential operational efficiency is determined when the flight planning tool is impacted by a loss of connectivity with a cloud-based device.

16. The method of claim 11, wherein the at least one operation of the aircraft comprises a navigation operation.

- 17.** The method of claim 11 further comprising: converting the potential operational efficiency to a value and provide the value to a user.
- 18.** The method of claim 17, further comprising: generating a bill, payable by the user, based on the value of the potential operational efficiency, wherein: the bill corresponds to a fraction of the value of the potential operational efficiency.
- 19.** The method of claim 11, further comprising: transmitting the potential operational efficiency to a cloud-based device while the aircraft is in flight.
- 20.** A non-transitory computer readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform a method comprising: providing at least one aircraft characteristic of one or more aircraft characteristics sensed by one or more sensors to a flight planning tool for processing by one or more features of the flight planning tool; receiving a recommended flight setting from the flight planning tool based on the processing by the flight planning tool; comparing the recommended flight setting with a current flight setting that is determined based on at least one of the one or more aircraft characteristics from the one or more sensors; determining a potential operational efficiency based on the comparison between the recommended flight setting and the current flight setting; and changing at least one operation of an aircraft based on the potential operational efficiency.
-